appear. However, if we instead set $\vartheta = \vartheta_2(\boldsymbol{q})$, then

$$\begin{aligned}\langle \vartheta_2(\boldsymbol{q}) f_1(\boldsymbol{q},\boldsymbol{p})\rangle &= \pm\frac{\beta}{2}\left\langle (\gamma U(\boldsymbol{q}) - c)\vartheta_2(\boldsymbol{q}) \pm \boldsymbol{p}^T \boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\vartheta_2(\boldsymbol{q})\right\rangle \\ &= \pm\frac{\gamma\beta}{2}\left\langle (U(\boldsymbol{q}) - \langle U(\boldsymbol{q})\rangle)\vartheta_2(\boldsymbol{q})\right\rangle,\end{aligned}$$

in which case our correction becomes $\mathcal{O}(\gamma)$. This is potentially disastrous if we choose the friction to be large (consider $\gamma \sim h^{-1}$), as the error introduced through discretization will dominate the true average to be sampled. Examining the algorithms themselves this is unsurprising, as although we have integrated the Ornstein-Uhlenbeck process exactly, when considering the limit of infinite friction in the ⟦XYZ⟧ schemes we obtain an inconsistent discretization. Hence these first-order schemes will be unsuitable for the integration of overdamped Langevin dynamics.

This is not always the case for first order schemes, however. For example the scheme denoted ⟦AOAB⟧ is consistent with the dynamics, and may be used for integrating the overdamped ($\gamma = \infty$) Langevin equation.

7.8.2 Numerical Experiment

Let us consider testing the computed first order corrections on a one dimensional example. Let $M = \beta = 1$ and define the biased double-well potential as

$$U(q) = (q^2 - 1)^2 + q/2.$$

We shall consider computing the average of a specific observable $\vartheta(q,p)$, and estimating the error due to discretization using the correction functions given in (7.25), where we choose the observable

$$\vartheta(q,p) = p^2 - qU'(q) + 2qp, \qquad \langle\vartheta\rangle = 0.$$

The low dimensionality of the problem and the simplicity of the observable and potential energy function make resolving the perfect sampling bias to leading order an easy task, allowing us to give a good estimate (to order h^2) of the discretization error. In Fig. 7.4 we plot the results of $\langle\vartheta\rangle_h$ computed via numerical experiment, using each of the six first order schemes at either $\gamma = 1$ or $\gamma = 2$, and compare them to our first order prediction

$$\langle\vartheta\rangle_h \approx \langle\vartheta\rangle + h\langle f_1\vartheta\rangle.$$

The numerical experiment at each timestep is averaged from 16 independent experiments, each running over a time interval of 10^8. The results match our

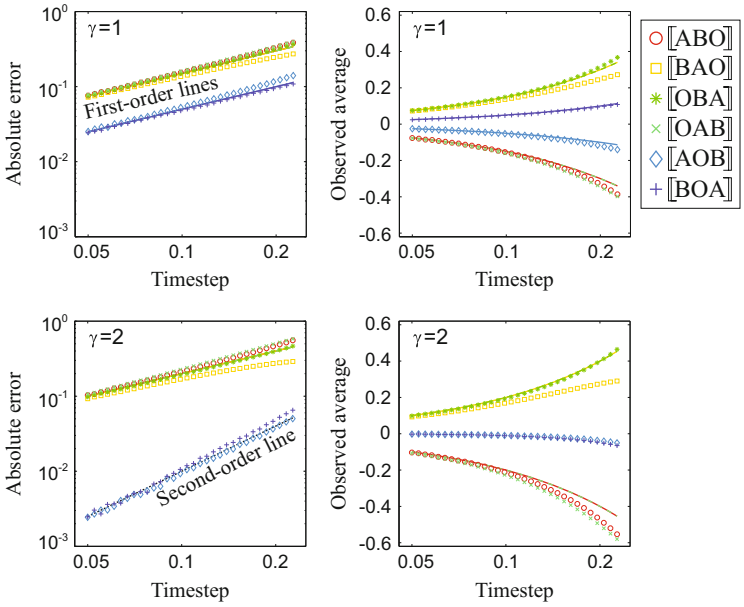


Fig. 7.4 We plot the absolute error in the computed value of $\langle \vartheta \rangle_h$ (*left*) and the observed value of $\langle \vartheta \rangle_h$ (*right*) as symbols, while our prediction computed using (7.25) is plotted as a *solid line*. For $\gamma = 1$ we find first order behavior (as expected), but for $\gamma = 2$ two methods exhibit higher order behavior. This is a special case for this observable that can be understood through our analysis

prediction extremely well, though at large stepsize there is evidence of higher order behavior $\mathcal{O}(h^2)$ having an effect on the computed average, evident by the divergence away from our prediction line.

The ⟦AOB⟧ and ⟦BOA⟧ schemes also demonstrate a surprising second order behavior for the experiments at $\gamma = 2$, that we would not necessarily expect. This is due to a felicitous choice of our observable, where the leading order correction can be shown to be zero. The fact that both of these schemes share this property is unsurprising, as $f_{1,[\![\mathrm{AOB}]\!]} = -f_{1,[\![\mathrm{BOA}]\!]}$, although for a general observable we would not expect second order averages for these methods. We could similarly construct special cases for other observables that would give higher order behavior for other schemes.

The first order correction term for the ⟦BOA⟧ scheme is given by the formula

$$\begin{aligned}\langle \vartheta f_{1,[\![\mathrm{BOA}]\!]} \rangle &= \frac{\beta}{2} \int_{\mathcal{D}} \left(p^2 - qU'(q) + 2qp\right)\left(pU'(q) + \gamma U(q) - c\right) \rho_\beta \mathrm{d}\omega, \\ &= \frac{1}{\beta} + \frac{\gamma}{2}\langle U \rangle - \frac{\gamma\beta}{2}\langle qUU' \rangle, \\ &= \frac{(2-\gamma)}{2\beta}.\end{aligned}$$

Hence for the critical value of $\gamma = 2$, the leading order correction will be zero, giving

$$\langle \vartheta \rangle_h^{[\![\mathrm{BOA}]\!]} = \langle \vartheta \rangle + \mathcal{O}(h^2).$$

This is exactly the behavior we see in Fig. 7.4.

7.9 Error Expansion for Symmetric Langevin Methods

Of particular interest for sampling the canonical distribution are symmetric Langevin methods. We believe these are likely to be the most useful class of methods for practitioners, as by symmetrizing the expansion the odd order terms in (7.16) vanish identically using the Jacobi identity in the BCH expansions. This implies that a symmetric scheme gives a second order error in computed averages. Many symmetric methods can be constructed that require only one evaluation of the force per iteration (effectively making them as inexpensive as a first order method).

In order to simplify the algebra, we will use $\boldsymbol{M} = \boldsymbol{I}$ throughout this section, which will not alter the major results and amounts to a linear change of variables. We shall consider general symmetric schemes of the form $[\![\mathrm{XYZYX}]\!]$, and perform a similar analysis to the previous section. The terms appearing in the expansions will be significantly more complex than those encountered in first order methods. We make use of the following proposition for symmetric compositions:

Proposition 7.1 *Let $\mathcal{L}_{\mathrm{X}}^\dagger$, $\mathcal{L}_{\mathrm{Y}}^\dagger$ and $\mathcal{L}_{\mathrm{Z}}^\dagger$ be general operators to which BCH can be applied, then defining*

$$e^{h\hat{\mathcal{L}}^\dagger} = e^{\frac{h}{2}\mathcal{L}_{\mathrm{X}}^\dagger} e^{\frac{h}{2}\mathcal{L}_{\mathrm{Y}}^\dagger} e^{h\mathcal{L}_{\mathrm{Z}}^\dagger} e^{\frac{h}{2}\mathcal{L}_{\mathrm{Y}}^\dagger} e^{\frac{h}{2}\mathcal{L}_{\mathrm{X}}^\dagger}, \tag{7.26}$$

gives

$$\hat{\mathcal{L}}^\dagger = \mathcal{L}_0^\dagger + h^2 \mathcal{L}_2^\dagger + \mathcal{O}(h^4),$$

where

$$\begin{aligned}
\mathcal{L}_0^\dagger =& \mathcal{L}_{\mathrm{X}}^\dagger + \mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger, \\
\mathcal{L}_2^\dagger =& \frac{1}{12}\left([\mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger] + [\mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{X}}^\dagger] \right) \\
& - \frac{1}{24}\left([\mathcal{L}_{\mathrm{Y}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger, \mathcal{L}_{\mathrm{Z}}^\dagger] + [\mathcal{L}_{\mathrm{X}}^\dagger, \mathcal{L}_{\mathrm{X}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger] \right).
\end{aligned}$$